



# UK TRAIN RIDES PROJECT

## Documentation

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# 1. Introduction and Overview:

## UK Train Rides Data Analysis

In an increasingly urbanized and mobile society, **efficient rail transportation** is critical to economic vitality and commuter satisfaction. The UK rail network, serving millions annually, is a complex ecosystem involving infrastructure, customer behavior, weather dynamics, and financial performance.

This analysis aims to provide a **comprehensive view of train operations**, focusing on three essential pillars

## Financial Performance and Ticketing Strategy

The *Revenue Analysis* dissects earnings across different ticket types, classes, and channels (online vs on-station). It also highlights the role of promotional programs like railcards and offers a path toward maximizing profitability while enhancing service accessibility.

## Passenger Behavior and Demand Patterns

The *Passenger Flow* section maps out when and where people are traveling. It reveals commuter peaks, popular departure hubs, and demand cycles based on time of day and week. This helps identify scheduling and staffing priorities.

## Operational Reliability

Through the *Delays and Cancellations* analysis, we explore the underlying causes of service disruptions, their effect on customer satisfaction, and financial implications such as refund volumes.

these insights support data-driven decisions to improve punctuality, manage capacity, enhance customer experience, and optimize revenue streams. This analysis is not only diagnostic, but also strategic, offering guidance for long-term planning and operational excellence in UK rail services.

## Objectives

This analytical project aims to achieve the following key goals to support operational efficiency, customer satisfaction, and financial optimization within the UK rail network:

### 1. Evaluate Revenue and Ticketing Performance

Break down revenue streams by ticket types, payment methods, and sales channels to identify growth opportunities and improve pricing strategies.

### 2. Understand Passenger Flow Patterns

Analyze when and where passengers travel across different times of day and days of the week to enable better scheduling and resource allocation.

### **3. Identify Causes of Delays and Cancellations**

Detect the most frequent and impactful reasons behind service disruptions, such as weather conditions or technical failures, to support service reliability improvements.

### **4. Enable Data-Driven Decision Making**

Empower stakeholders with interactive dashboards and actionable insights to continuously monitor performance and enhance service delivery.

### **Target Audience:**

This project is designed to serve multiple stakeholder groups within the railway ecosystem:

#### **1. Train Operating Companies**

To enhance operational performance, reduce inefficiencies, and improve customer experience based on real-time insights.

#### **2. Regulatory and Government Bodies**

Such as the Department for Transport and rail oversight authorities, for policymaking, compliance monitoring, and long-term planning.

#### **3. Data Analytics and Planning Teams**

Within transportation organizations, to support strategic and operational decision-making using evidence-based analysis.

#### **4. Investors and Transport Sector Stakeholders**

To evaluate revenue performance, growth potential, and return on investment in the UK rail infrastructure.

#### **5. Users (Passengers)**

While not directly involved in the analytics process, the outcomes aim to improve their travel experience by reducing delays and optimizing ticket options.

## 2. Data Source and Description:

The data used in this project was provided as part of a training initiative under the Digital Egypt Pioneers (DEPI) program, in collaboration with the training provider CLS (Creative Learning Solutions). The sources include:

### Railway csv file:

UK trains dataset contains train ticket data for Rail services in the United Kingdom in 2023/2024.

**Rows:** 31,653

**Columns:** 18

### Key Attributes:

| Field                      | Description                                                                                                                                                                                                                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Transaction ID</b>      | Unique identifier for an individual train ticket purchase.                                                                                                                                                                                                                                                                |
| <b>Date of Purchase</b>    | Date the ticket was purchased.                                                                                                                                                                                                                                                                                            |
| <b>Time of Purchase</b>    | Time the ticket was purchased.                                                                                                                                                                                                                                                                                            |
| <b>Purchase Type</b>       | Whether the ticket was purchased online or directly at a train station.                                                                                                                                                                                                                                                   |
| <b>Payment Method</b>      | Payment method used to purchase the ticket (Contactless, Credit Card, or Debit Card).                                                                                                                                                                                                                                     |
| <b>Railcard</b>            | Whether the passenger is a National Railcard holder (Adult, Senior, or Disabled) or not (None). Railcard holders get 1/3 off their ticket purchases.                                                                                                                                                                      |
| <b>Ticket Class</b>        | Seat class for the ticket (Standard or First)                                                                                                                                                                                                                                                                             |
| <b>Ticket Type</b>         | When you bought or can use the ticket. Advance tickets are 1/2 off and must be purchased at least a day prior to departure. Off-Peak tickets are 1/4 off and must be used outside of peak hours (weekdays between 6-8am and 4-6pm). Anytime tickets are full price and can be bought and used at any time during the day. |
| <b>Price</b>               | Final cost of the ticket.                                                                                                                                                                                                                                                                                                 |
| <b>Departure Station</b>   | Station to board the train.                                                                                                                                                                                                                                                                                               |
| <b>Arrival Destination</b> | Station to exit the train.                                                                                                                                                                                                                                                                                                |
| <b>Date of Journey</b>     | Date the train departed.                                                                                                                                                                                                                                                                                                  |
| <b>Departure Time</b>      | Time the train departed.                                                                                                                                                                                                                                                                                                  |
| <b>Arrival Time</b>        | Time the train was scheduled to arrive at its destination (can be on the day after departure)                                                                                                                                                                                                                             |
| <b>Actual Arrival Time</b> | Time the train arrived at its destination (can be on the day after departure).                                                                                                                                                                                                                                            |
| <b>Journey Status</b>      | Whether the train was on time, delayed, or cancelled                                                                                                                                                                                                                                                                      |
| <b>Reason for Delay</b>    | Reason for the delay or cancellation.                                                                                                                                                                                                                                                                                     |
| <b>Refund Request</b>      | Whether the passenger requested a refund after a delay or cancellation.                                                                                                                                                                                                                                                   |

**Date table:**

To enhance the depth of time-based analysis, a Date Table was created and linked to the dataset. This table allowed for more advanced time intelligence calculations, such as:

- Day, Week, Month, Quarter, and Year breakdowns
- Day of Week and Weekend/Weekday identification
- Holiday and Season tagging (if applicable)
- Fiscal Periods for financial trend analysis

This rich dataset provides a robust foundation for analyzing customer behavior, revenue performance, and service reliability across the rail network, enabling insights that mimic real-world transport analytics scenarios.

**Holiday Table:**

Contains information about public holidays:

- Date: Holiday date.
- Day: Day of the week or day number.
- Holiday: Name or type of the holiday.

**Time of day Table:**

A dimension table used to categorize time into segments (e.g., morning, afternoon, evening):

- Time of Day: Text label for time segment.
- Time Order: Numeric value for sorting purposes.

### 3. Data Analysis questions:

#### Revenue & Sales Analysis

- What is the total revenue generated by ticket type and class?
- How do online and in-station purchases compare in volume and revenue?
- Which ticket types (e.g., Advance, Anytime, Off-Peak) are most popular and profitable?
- What is the average fare per journey, and how does it vary by class or route?

#### Passenger Behavior

- What are the peak travel hours and days of the week?
- Which departure stations have the highest passenger volumes?
- How does passenger demand vary by time, day, and season?
- Do certain customer segments (e.g., students, seniors) prefer specific time slots?

#### Operational Performance

- What are the main causes of train delays and cancellations?
- Which routes or time slots experience the most frequent disruptions?
- How do delays affect refund volumes and customer compensation?
- Are there patterns between weather conditions and delay frequency?

#### Ticketing Trends

- How often are railcards used, and how do they impact revenue?
- Which payment methods are most used?
- Is there a correlation between ticket type and likelihood of delay or refund?

#### Time-Based Insights

- How do sales and cancellations vary across months, quarters, and years?
- Are there seasonal trends in travel patterns or revenue?
- Do holidays or weekends significantly impact ticket purchases or delays?

## 4. Data Cleaning and preparation Process:

Data cleaning was conducted using Power Query to ensure consistency and remove any irrelevant information. Below are the detailed steps performed in figure 1:

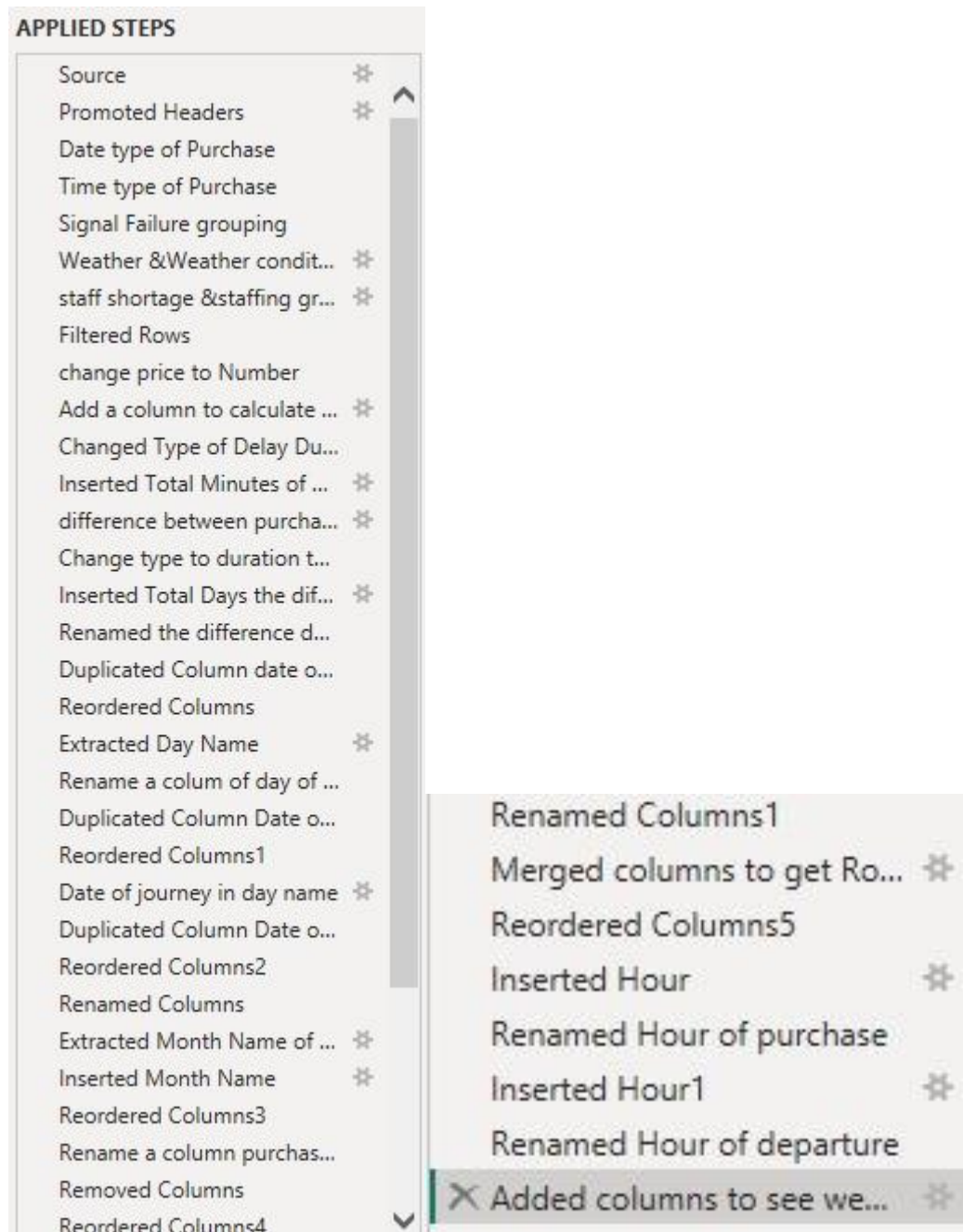


Figure 1 Applied steps in power query editor

## 5. Data Modeling:

In this project, a star schema was implemented to efficiently organize the data for reporting and analysis purposes. The model is designed to analyze and visualize railway journey data by integrating various related dimensions such as date, time of day, and public holidays. The model consists of a central fact table (railway) that records detailed information about each journey, including departure and arrival times, dates of travel and purchase, and station details.

To enhance analytical capabilities, the model includes supporting dimension tables:

- A **Date Table** to enable time-based filtering and grouping (e.g., by day, month, weekday).
- A **Holiday Table** to distinguish between regular days and public holidays.
- A **Time of Day Table** to categorize trips into segments like morning, afternoon, and evening.
- A dedicated **Measures Table** (All Measures) to store and manage DAX calculations used in reports and dashboards.

### Relationships Overview:

- All visible relationships are **one-to-many (1:\*)**, where dimension tables (like Date, Holiday, TimeofdayTable) act as lookup tables.
- The **railway** table serves as the fact table with repeated date and time values, allowing flexible slicing and analysis.

Through the established relationships between these tables (look Figure 2), users can gain insights into travel behavior trends, holiday impacts, and time-based patterns in a flexible and scalable way.



## UK TRAIN RIDES PROJECT

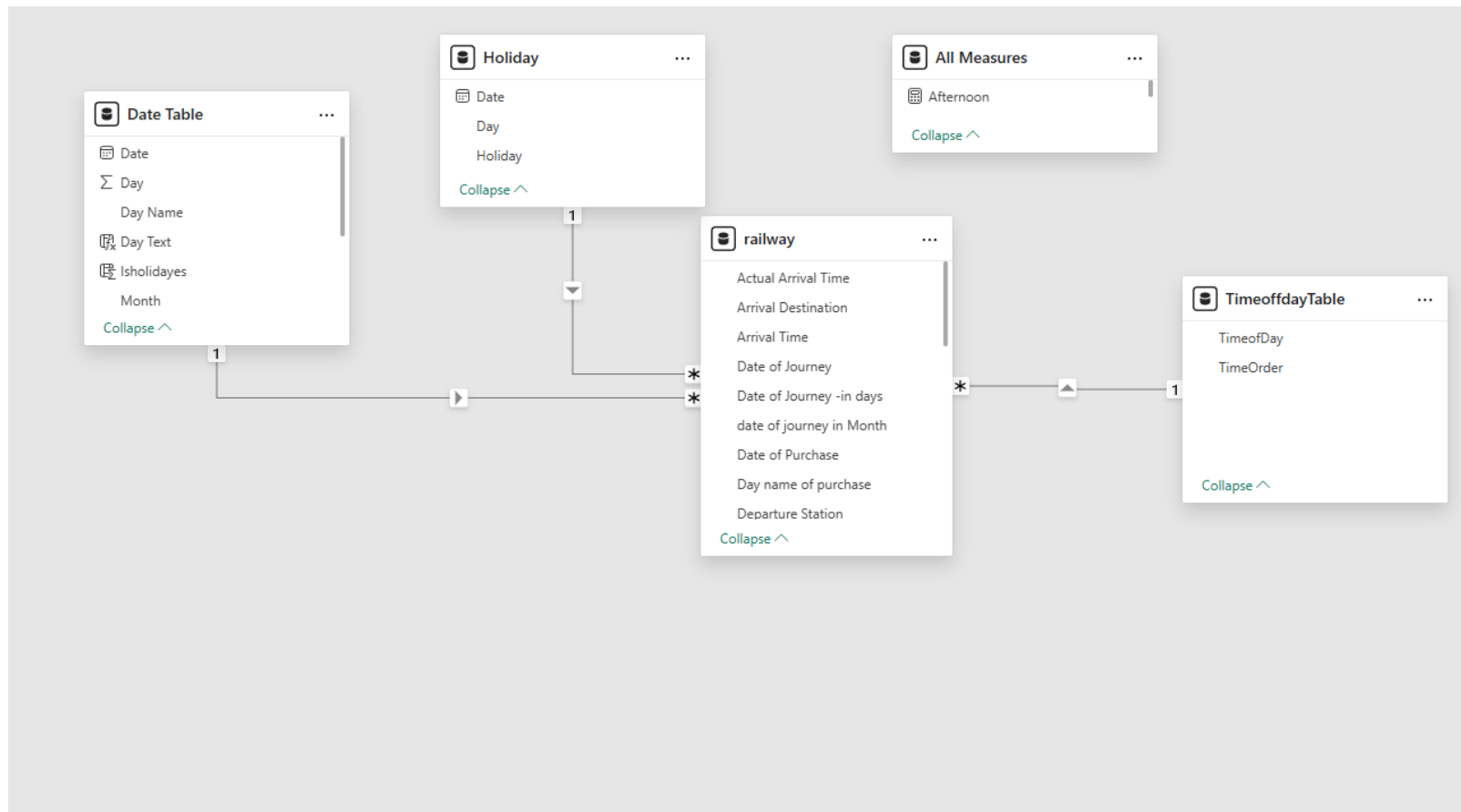


Figure 2 Data Modeling

## 6. Analysis Methodology:

### DAX:

These are DAX measures stored in a dedicated Measures Table called All Measures. They likely use DAX functions such as:

SUM(), AVERAGE(), COUNT(), CALCULATE(), DIVIDE(), FILTER(), etc.

| Ticket Price & Revenue Measures |                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------|
| Average Ticket Price            | Calculates the average price of sold tickets.                                                 |
| Max Price, Min Price            | Maximum and minimum ticket prices.                                                            |
| Total Tickets Sold              | Total number of tickets sold.                                                                 |
| Total Revenue                   | Total revenue from ticket sales                                                               |
| Total Refund                    | Total amount of refunded money.                                                               |
| Refund request                  | Number of refund requests submitted.                                                          |
| Refund percentage               | Ratio of refunded tickets to total sold tickets                                               |
| Passenger-Related Measures      |                                                                                               |
| Total Passengers                | Total number of passengers.                                                                   |
| Percentage Passengers           | May represent passenger load percentage or share of passengers per trip or time segment.      |
| Journey & Route Metrics         |                                                                                               |
| Total Journies                  | Total number of train journeys made.                                                          |
| Total Routes, Number of Routes  | Number of distinct routes traveled (may be duplicates or variations of the same calculation). |
| Delays & Cancellations          |                                                                                               |
| Total cancelled                 | Number of trips that were cancelled.                                                          |
| Cancelled Percentage            | Cancelled trips as a percentage of total trips.                                               |
| Delay Percentage                | Ratio of delayed trips to total trips.                                                        |
| Average Delay                   | Average delay time across all trips.                                                          |
| Total Delay                     | Sum of all delay times.                                                                       |
| Booking                         |                                                                                               |
| On Station                      | Likely counts bookings made at the station.                                                   |
| Online                          | Bookings made through an online platform.                                                     |

## 7. Dashboard Creation:

### Dashboard Design:

used alongside pre-made icons from reputable design websites (e.g., Flaticon, Iconfinder). These elements were integrated to create an intuitive, user-friendly interface that supports decision-making.

The dashboards were styled using a consistent and professional color palette (look Figure 3), ensuring visual harmony and effective data storytelling. The chosen palette emphasizes clarity, contrast, and accessibility—aligning with modern UI/UX best practices for analytical reports.



*Figure 3 Color Palette used in dashboards*

The data was impressively displayed with three dashboards that were made up:

(1) **Revenue and Ticketing Analysis (look Figure 4):**

**Financial Metrics:**

- **Total Revenue:** \$742K
- **Average Ticket Price:** \$23.44
- **Total Tickets Sold:** 32K

**Insights:**

- **Revenue by Card Type:**

Most revenue came from passengers without a railcard (\$0.57M), with significantly less from discounted categories.

- **Top Routes by Revenue:**

**London Kings Cross → York** is the top-earning route (\$183K), followed by **Liverpool Lime Street → Manchester** (\$113K).

- **Purchase Behavior:**

**Online purchases (383K)** slightly exceed **on-station sales (359K)**.

**Credit Cards** dominate across both online and station sales.

- **Ticket Type Breakdown:**

**Advance tickets** generate the most revenue (\$242K), followed by **Off-Peak** and **Anytime**.

# UK TRAIN RIDES PROJECT

Explore trends in ticket prices and revenue performance within the UK rail network

## UK Train Rides Analysis



Figure 4 Revenue Analysis

## (2) Passenger Flow Analysis (Look Figure 5)

### Time-based Passenger Distribution:

- **Morning:** 37% of traffic
- **Evening & Afternoon:** 24% & 20%
- **Night:** 19%

### Insights:

- **Peak Departure Hours:**

**7 AM and 6 PM** have the highest departure activity (up to 3,113 passengers).
- **Passenger Distribution Over Days:**

High activity on **Fridays and weekends**, which coincide with increased train load.
- **Top Departure Stations:**

**Manchester Piccadilly** (5,650 passengers), **London Euston** (4,954), and **Liverpool Lime Street** (4,561).
- **Trends Over Time:**

Passenger counts remained stable, with slight dips possibly around holidays or service disruptions.

# UK TRAIN RIDES PROJECT

Explore passenger flow patterns and station activity across different times of the day

## UK Train Rides Analysis

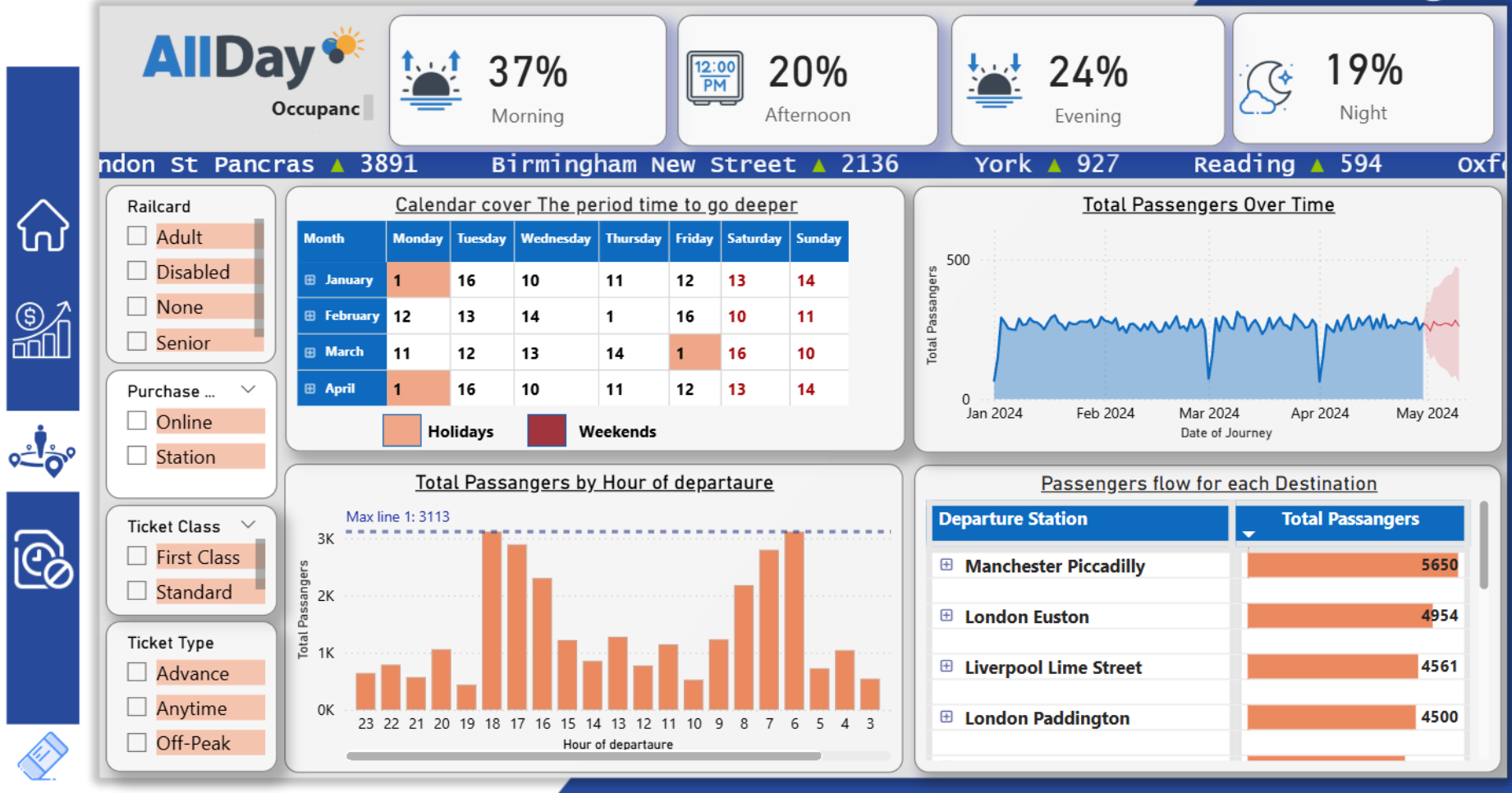


Figure 5 Flow Analysis

### (3) Delays and Cancellations Analysis (Look Figure 6)

#### Key Metrics:

- **Total Passengers:** 2,292
- **Total Refund:** £26K
- **Refund Requests:** 546
- **Average Delay:** 42.21 minutes

#### Insights:

- **Primary Cause of Delays:**

**Weather** is the top cause (40.45%), followed by technical issues (20.59%) and signal failure (19.68%).

- **Refunds Behavior:**

76.18% of affected passengers did **not** request a refund, while 23.82% did.

- **Busiest Route (by occupancy):**

**Liverpool Lime Street** dominates with 34.03% of total occupancy.

- **Passenger Patterns:**

Delays and passenger numbers vary over time, with consistent fluctuations from Jan–April 2024.



# UK TRAIN RIDES PROJECT

Explore key trends and factors contributing to delays and cancellations in UK train services

## UK Train Rides Analysis

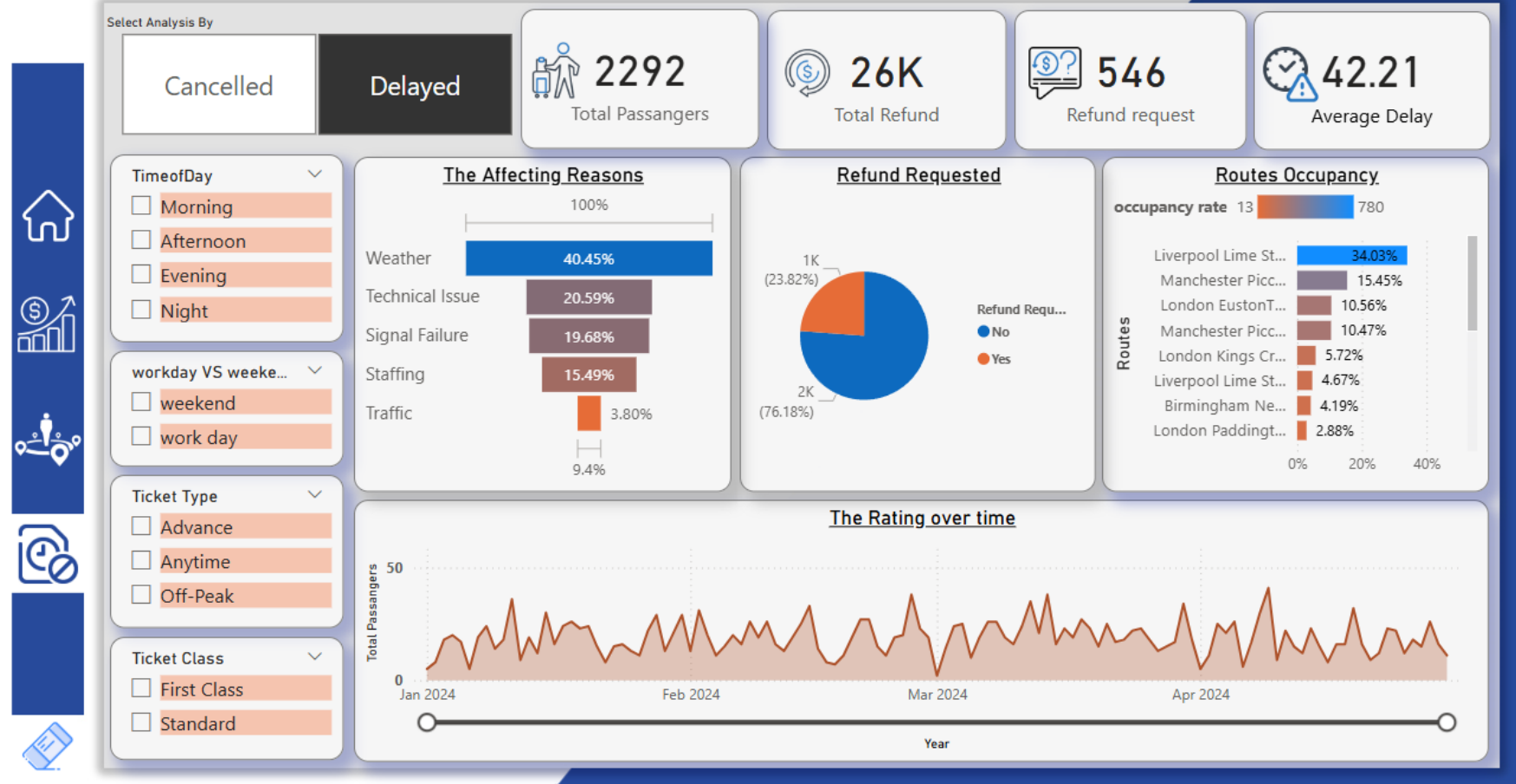


Figure 6 Delays and Cancellations Analysis

## 8. Recommendations

### I. Improving Operational Efficiency & Performance Management

- **Delay Reduction & On-Time Performance:**
  - Perform root cause analysis to identify delay drivers.
  - Invest in infrastructure and optimize scheduling.
  - Improve incident management and staff training.
  - Implement preventive maintenance.
- **KPI Management & Monitoring:**
  - Define and track key performance indicators (KPIs).
  - Set performance targets and report on performance.
- **Disruption Management:**
  - Minimize delayed and cancelled trips.
  - Improve passenger communication and contingency planning.
- **Impact Analysis & Trend Monitoring:**
  - Analyze the impact of delays on revenue and passenger satisfaction.
  - Monitor trends in delays, passenger numbers, and overall performance.
  - Use predictive modeling and real-time monitoring.
  - Conduct longitudinal studies and consider external factors.
- **Station & Route Optimization:**
  - Assess station capacity and optimize routes.
  - Improve service planning and intermodal integration.
- **Refund Management:**
  - Analyze and optimize refund processes and policies.
  - Gather and analyze customer feedback on disruptions and refunds.

### II. Optimizing Revenue

- **Revenue Optimization:**

- Analyze revenue drivers and forecast demand.
- Optimize pricing strategies using price elasticity and dynamic pricing.
- Implement yield management.
- **Sales Enhancement:**
  - Boost ticket sales through marketing and sales channel optimization.
  - Develop partnerships and customer loyalty programs.
- **Customer Segmentation:**

Analyze revenue by customer segments and tailor pricing, marketing, and services accordingly.
- **Payment Optimization:**
  - Optimize payment methods by diversifying options, reducing transaction costs, and enhancing security.
  - Promote contactless payments.
- **Revenue & Route Management:**
  - Track revenue trends and forecast future revenue.
  - Analyze route profitability and optimize resource allocation

## 9. Conclusion

This report provides a comprehensive overview of UK rail network performance. The outlined recommendations, focusing on operational efficiency, revenue optimization, and passenger experience, offer a roadmap for positive change. Implementing these strategies will help mitigate disruptions, maximize revenue, and improve passenger satisfaction.